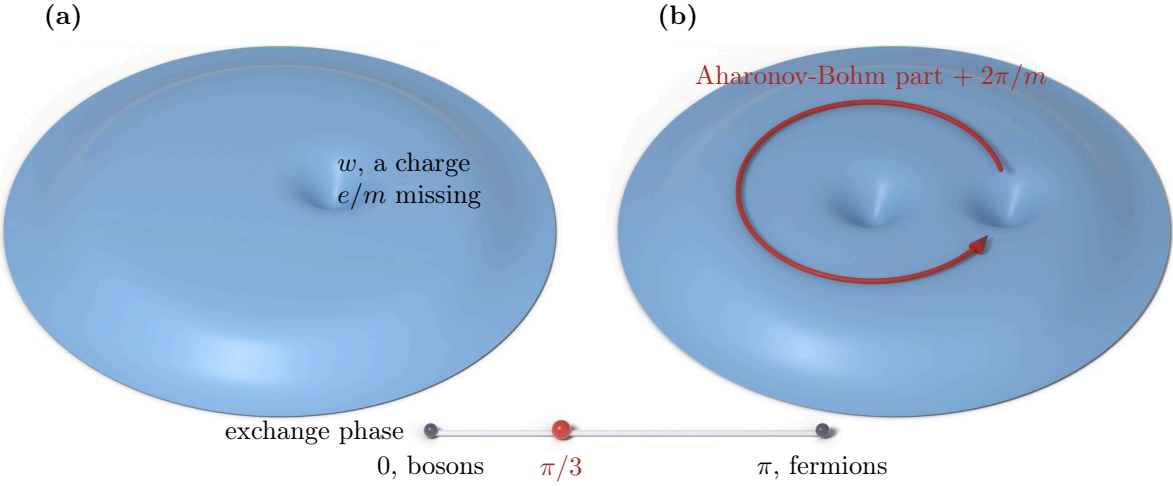

(a)
(b)
Aharonov-Bohm part + $2\pi/m$
w, a charge
e/m missing
exchange phase
0, bosons
$\pi/3$
π, fermions